

Customer Churn Prediction System

SYNOPSIS

Project Title

Customer Churn Prediction System Using Multi-Agent Machine Learning Architecture

Objective

To build an intelligent system that predicts which telecom customers are likely to leave (churn) and provides actionable retention recommendations using a multi-agent machine learning architecture.

Problem Statement

Telecom companies lose significant revenue when customers leave. Identifying at-risk customers early allows companies to take proactive retention measures. This project automates the prediction process using AI agents.

Solution Overview

The project implements a 3-agent architecture:

- 1. DataPrepAgent - Handles data cleaning and preprocessing
- 2. PredictionAgent - Trains ML models and makes predictions
- 3. RecommendationAgent - Generates retention strategies

A Streamlit web dashboard provides an interactive interface for users to input customer data, view predictions, analyze trends, and upload bulk CSV files.

Technologies Used

Category	Technology
Language	Python 3.8+
Machine Learning	Scikit-learn, XGBoost
Data Processing	Pandas, NumPy
Web Framework	Streamlit
Visualization	Plotly
Model Persistence	Joblib

ML Models Implemented

Customer Churn Prediction System

Model	Type	Accuracy
Logistic Regression	Linear	79.48%
Random Forest	Ensemble	78.69%
XGBoost	Gradient Boosting	78.20%

Dataset

- **Source**: Telco Customer Churn Dataset
 - **Records**: 5,043 customers
 - **Features**: 20 columns (demographics, services, billing, churn)
 - **Target**: Churn (Yes/No)
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Features

1. Single customer prediction with risk gauge
 2. Interactive data analytics charts
 3. Model performance comparison
 4. Bulk CSV upload and analysis
 5. Downloadable results
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Applications

- Telecom customer retention
 - Subscription service management
 - SaaS churn reduction
 - Banking customer loyalty programs
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Future Scope

- Real-time API integration
 - Deep learning models (LSTM, Neural Networks)
 - Customer segmentation clustering
 - Automated email/SMS alerts
 - A/B testing for retention strategies
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Conclusion

The system successfully predicts customer churn with ~79% accuracy using a modular multi-agent architecture, enabling data-driven retention decisions.

References

Customer Churn Prediction System

1. IBM Telco Customer Churn Dataset
2. Scikit-learn Documentation
3. Streamlit Documentation
4. XGBoost Documentation